

3D & Spatial Visualizations

Presentation boards and illustrative views. Renders are artistic impressions; the analysis outputs are computed.

CONTENTS

- Board 1 concept
- Board 2 evidence
- Fig10 masterplan

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository github.com/wasim437/AI_SAFA
Live portal wasim437.github.io/AI_SAFA/
Produced by [src/boards.py](#) · [src/plan.py](#) · [archive/withdrawn_visuals/README.md](#)
Reproduce `python run_analysis.py` · `python -m tests.test_pipeline`

How this document was produced

This submission is the output of a ten-phase process in which each phase is checked against the one before it. Nothing downstream is hand-drawn or hand-typed: change an input, re-run, and every chart, drawing and figure moves with it — or a test fails loudly. The phases behind this particular document are marked.

Phase 1 — Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

Phase 2 — Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

Phase 3 — Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

Phase 4 — Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

Phase 5 — Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

Phase 6 — Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

Phase 7 — Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

Phase 8 — User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

Phase 9 — AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

Phase 10 — Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

THE CODE AND DATA BEHIND THIS DOCUMENT

[src/boards.py](#)

[src/plan.py](#)

[archive/withdrawn_visuals/README.md](#)

REPRODUCE IT

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 38 checks
```

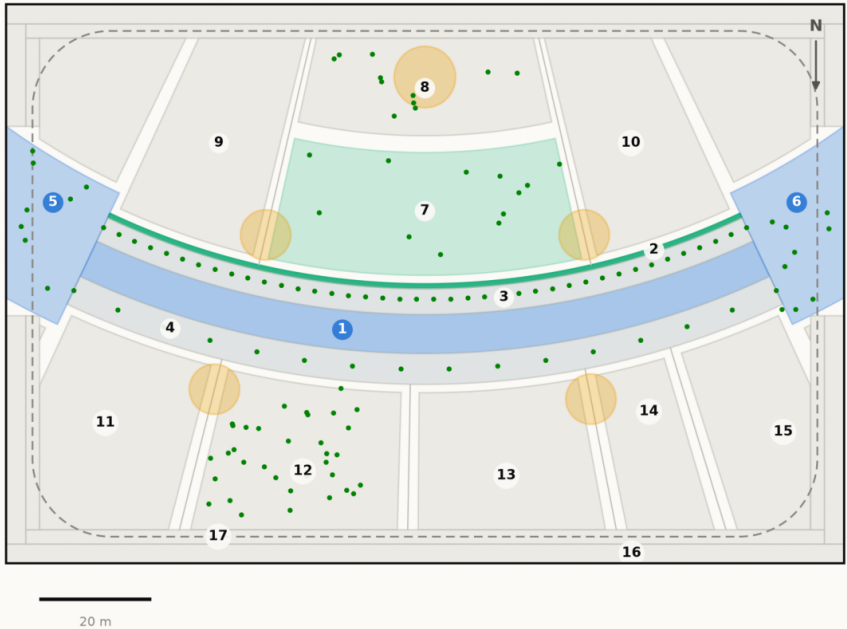
AL SAFA 2 PARK — FALAJ AL SAFA

Dubai Municipality AI Park Design Challenge | Board 1 of 2 — CONCEPT

The concept, the plan, and the section that makes it work

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150x100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

ROOM SCHEDULE

1	Al Mamsha — the Crescent Walk	871 m²
2	Al Falaj — the water channel	105 m²
3	Crescent Shade Margin (N)	549 m²
4	Crescent Shade Margin (S)	715 m²
5	West Gate Majlis	439 m²
6	East Gate Majlis	439 m²
7	Al Nakhil — the Oasis Basin	1,140 m²
8	Quiet Contemplation Garden	707 m²
9	Children's Dune Play	1,267 m²
10	Family Picnic Grove	1,267 m²
11	Outdoor Fitness Terrace	908 m²
12	Native Planting / Biodiversity Wadi	894 m²
13	Community Plaza & Event Lawn	787 m²
14	Souk Kiosks & Services	417 m²
15	Multipurpose Sports Lawn	630 m²
16	Al Kathib — the dune berm	1,464 m²
17	Al Madar — the perimeter loop	986 m²
	Al Sikkak — shaded alleys & setbacks	1,414 m²
Total		15,000 m²

The crescent from the north-west, at golden hour

Visualisation in preparation

Aerial view · AL_SAFa_MASTER_PROMPT.md prompt 01

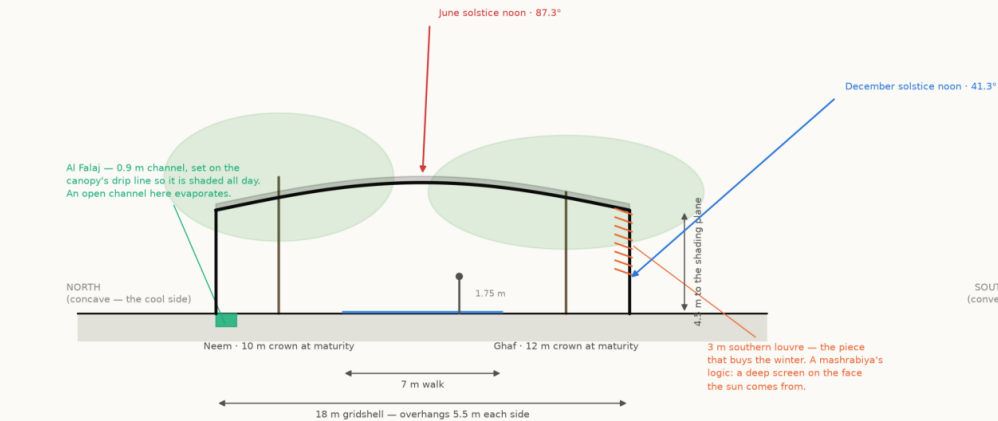
Beneath Al Hilal — the perforated soffit at eye level

Visualisation in preparation

Eye-level view · AL_SAFa_MASTER_PROMPT.md prompt 03

Section A-A — the Crescent Canopy at midspan

Every dimension read from config.CRESCENT; sun angles computed by the NREL algorithm at 25.190°N

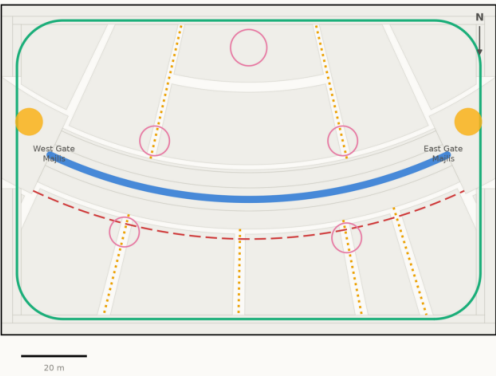


The plane alone loses the walk to a low southern sun. The louvre is what holds the shadow on the path from November to January.
Source: src/config.py CRESCENT; sun angles NREL SPA via pvlib · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

One arc. Every room in the park is struck off its centre, so no room is a rectangle and every room faces the crescent square-on.
The bow is 18 m deep on a 141 m radius — swept against the 8,760-hour solar model, not drawn by eye. A straight canopy shades marginally more ground on average and leaves the walk with no shade at all for 330 hours a year; this one leaves it for 56.
Walk shaded 87.3% of daylight hours · 131 trees · 15,000 m² · every number reproducible with `python run\_analysis.py`

Circulation & accessibility

One shaded primary route, radial alleys off it, and a running loop around the whole



Al Mamsha — primary route, shaded, step-free
Al Sikkak — shaded alleys to the rooms
Al Madar — 438 m running loop
Service / emergency vehicle access
Majlis — shaded rest, never more than 33 m from the walk

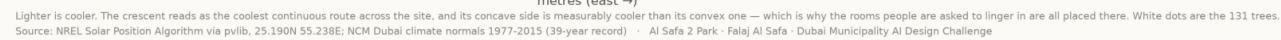
Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped.
Source: srcplan.py — routes are the drawn geometry, not a sketch over it · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Presentation board.

Dubai Municipality AI Park Design Challenge | Board 2 of 2 — EVIDENCE

Predicted summer comfort across the site

July afternoon heat index per square metre — the shade strategy, tested



What actually determines whether a square metre is shaded

Permutation importance — the drop in accuracy when each feature is shuffled



Planting plan – 131 trees

Drawn at mature canopy radius. The avenue is structural; it carries the shoulder seasons the gridshell cannot



28 of 131 trees are UAE natives (Ghaf and Date Palm). Peak summer irrigation demand 11.8 m³/day. Ghaf takes the southern rank because it is the most drought-tolerant species in the schedule and that rank is the more exposed of the two. Source: data4wsppecies, water, carbon, rates.csv; layout from sc4shelery - Al Safa 2 Park - Fala Al Safa - Dubai Municipality Al Design Challenge

Shade converts unusable daylight hours into usable ones

Share of the year's 4,402 daylight hours in each comfort band



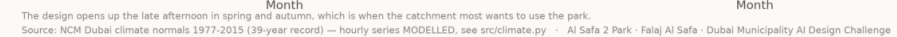
Annual shade coverage by zone type

Ray-traced ground-plane occlusion, 1 m grid, sampled across the year



When is the park comfortable?

Mean heat index by hour and month — exposed today, shaded as designed



Elevation — the Crescent Canopy
60 m of the 124 m run, to scale, 6 m structural bay

60 m of the 124 m run, to scale, 6 m structural bay



Comfortable daylight hours rise from 44.5% to 64.6% — 20.1 points of the year handed back. Peak heat index falls 56.8 °C → 48.7 °C.

Site-wide mean shade is 34.1%, and that is stated as a position rather than hidden: this scheme makes a few places genuinely excellent instead of the whole site marginally better.

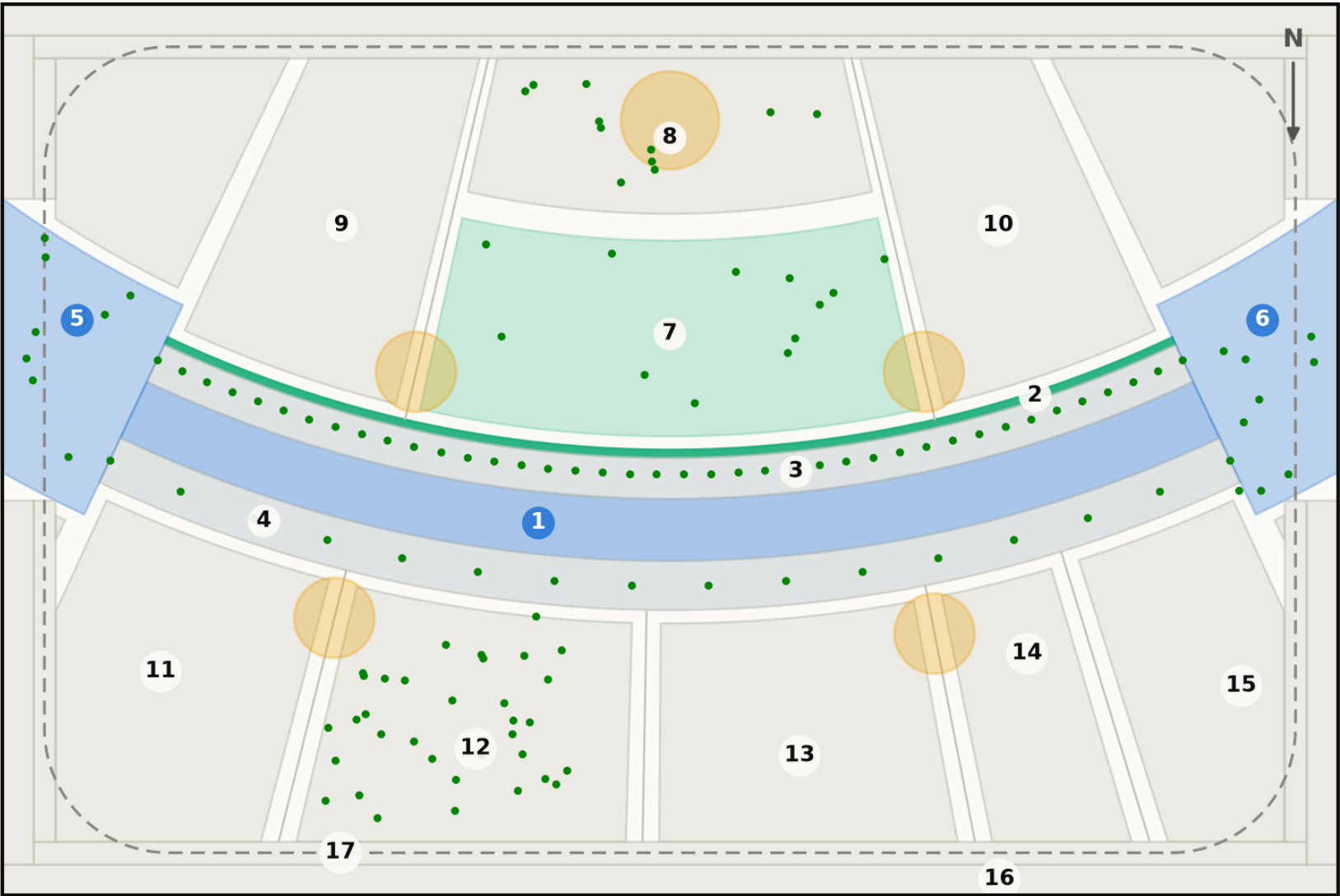
Shade surrogate R^2 0.994 · comfort classifier 97.5% with temperature and humidity withheld · 2 microclimate regimes, k selected by silhouette

Fig10 masterplan

Technical drawing — to scale. Geometry generated by src/plan.py; every area is the measured area of the drawn polygon.

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



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Total 15,000 m²

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Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

The complete project, and where to check it

Every link below is live. The repository holds the data, the code, the trained models and the tests, and runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository github.com/wasim437/AI_SAFA

Live portal wasim437.github.io/AI_SAFA/

START HERE

EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language
PROJECT_PLAN.md	Requirements, phases, status, what is left
README.md	The design argument, written for a juror
notebooks/AL_SAFA_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded

THE GEOMETRY AND THE MODELS

src/plan.py	Single source of the crescent geometry
src/climate.py	The 8,760-hour year rebuilt from NCM normals
src/solar.py	Sun position and shadow ray-tracing
src/dataset.py	Assembles the ML training tables
src/models.py	The four models
src/costing.py	The cost model against the AED 35 M ceiling

THE DATA, AS ISSUED AND AS PROCESSED

data/raw/sources.json	Every source dataset, with its period
DATA_SOURCES.md	Sources and their stated limitations
data/raw/site_zoning_schedule.csv	The measured room schedule
data/processed/hourly_climate_comfort_8760.csv	The 8,760-hour climate and comfort series
data/processed/cost_plan.csv	The capital cost plan, line by line

THE RESULTS, AND THE CHECKS ON THEM

models/model_metrics.json	Trained-model metrics
models/headline_metrics.json	The headline numbers
tests/test_pipeline.py	38 correctness checks
archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why

THE DRAWINGS

figures/fig10_masterplan.png	Masterplan and room schedule
design/visuals/section_crescent.png	Section A–A at midspan
design/visuals/circulation_crescent.png	Circulation and accessibility
design/visuals/facilities_crescent.png	Commercial & Service Facilities Map
design/visuals/planting_crescent.png	Planting plan, 131 trees